Sumi Suresh M.S., Ph.D.

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Professional Summary

AI and machine learning researcher, educator, and former software engineer with experience building computer vision, handwriting recognition, document intelligence, and multimodal systems. Develops data, training, evaluation, and inference pipelines in Python and PyTorch, including transformer and vision-language model fine-tuning on NVIDIA GPUs. Current work on NSF- and Google-funded projects converts child-handwriting data into recognition, screening, and progress-monitoring tools for education. Skilled in deep learning, natural language processing, and computer vision, with a strong publication record in leading journals and conferences.

Skills

Programming	Python, Matlab, Java/J2EE, ASP.NET, C, C++
Deep Learning	PyTorch, TensorFlow, Keras
Models and Methods	CNNs, BiLSTMs, Transformer architectures, LoRA, QLoRA
Computer Vision and Document AI	OpenCV, OCR and handwriting text recognition, image processing, segmentation, feature extraction, multimodal learning
GPU and Systems	CUDA, NVIDIA A100 GPUs, single- and multi-GPU training, FSDP, model inference, performance profiling, Linux
Development	Visual Studio Code, Eclipse, data preprocessing, experiment design, model evaluation, technical documentation
Research & Academic	Scientific writing, peer review, mentoring, teaching
Other	Research design, model development, data analysis, interdisciplinary collaboration

Employment History

Nov 2023 – Present — **Postdoctoral Research Associate,** Dept. of Computer Science and Engineering, State University of New York (SUNY) at Buffalo, Buffalo, USA.

Aug 2022 – Sep 2023 — **Assistant Professor,** Dept. of Computer Science and Engineering, School of Computing, Amrita Vishwa Vidyapeetham, Amritapuri Campus, Kerala, India.

Sep 2011 – Aug 2012 — **Software Engineer (Java Developer).** Ospyn Technologies, Thiruvananthapuram, Kerala, India.

Education

2017 – 2022 — **Ph.D. Computer Science**, Amrita Vishwa Vidyapeetham, India.
Thesis title: *Novel Approaches To Optimize Camera Coverage in Multi-Camera Surveillance Networks.*

2014 – 2016 — **M.Tech. Computer Science and Engineering (Image Processing)**, Cochin University of Science and Technology (CUSAT), India.
Thesis title: *Lung Cancer Detection System Using Genetic Algorithm-Based Feature Selection on Sputum Cytology Images.*

2007 – 2011 — **B.Tech. Computer Science and Engineering**, Cochin University of Science and Technology (CUSAT), India.

Projects and Research

- **AI for Progress Monitoring & Goal Personalization in IEPs for Students with Writing Disabilities**
  ***Technologies**: Python, PyTorch, OpenCV, NLP, handwriting analysis, desktop application development*
  Developing a Google-funded AI toolkit at the University at Buffalo that analyzes scanned and tablet-based student handwriting. The toolkit combines handwriting measures with text-based analysis to support progress monitoring, personalized IEP goals, growth charts, and editable teacher-ready reports.

- **AI-based Screening of Dysgraphia and Dyslexia in Children Using Handwriting Analysis**
  ***Technologies**: PyTorch, Transformers, LoRA, QLoRA, FSDP, CUDA, multimodal learning*
  Contributed to this NSF-funded project under the National AI Institute for Exceptional Education, a multi-university research initiative in the United States. Developed deep-learning and document-analysis methods for recognizing and analyzing child handwriting to support the early screening of dyslexia and dysgraphia. Built handwriting recognition models that generate both faithful and intended-text transcriptions, including explicit handling of spelling errors and character reversals. Created reproducible vision-language model training and inference pipelines with dataset validation, checkpoint and adapter management, and single- and multi-GPU experiments.

- **Medical Image Analysis for Stroke Detection**
  ***Technologies**: Deep learning, generative models, image segmentation, medical image processing*
  Developed deep-learning methods for ischemic-stroke assessment from non-contrast CT images. Worked on generative models for producing perfusion-related maps directly from non-contrast CT and transformer-guided segmentation methods for identifying infarct-core and affected brain regions.

- **Novel Approaches to Optimize Camera Coverage in a Multi-Camera Surveillance Network**
  ***Technologies**: Algorithms, optimization, computer vision, surveillance systems*
  Developed scalable optimization methods for maximizing camera coverage under constraints related to camera availability, coverage requirements, energy consumption, and camera failures. The work included camera placement, energy-aware operation, failure recovery, and surveillance-network reconfiguration for indoor and outdoor environments.

- **Automated Lung Cancer Detection using Sputum Cytology Images**
  ***Technologies**: Image processing, feature extraction, genetic algorithms, classification*
  Developed an automated lung-cancer detection system in collaboration with the Regional Cancer Center, India. Designed methods to identify relevant features from sputum cytology images and improve classification performance.

Patents and Research Publications

Patents

1. **M. S. S. Suresh**, S. Renjith, and R. Manazhy, "Method for generating a pattern sequence to represent a sign language gesture classes," 202 441 096 451, Application awaiting examination, Dec. 6, 2024.
2. **M. S. S. Suresh** and V. Menon, "Method for camera coverage optimization under energy constraints," 202 441 068 354, Application awaiting examination, Sep. 10, 2024.
3. **M. S. S. Suresh** and V. Menon, "System and method for efficient reconfiguration of surveillance networks," 202 241 007 289, Reply Filed. Application in amended examination, Feb. 11, 2022.

4. **M. S. S. Suresh** and V. Menon, "System and methods for designing an optimal multi-camera surveillance network," 202 141 022 565, Reply Filed. Application in amended examination, May 20, 2021.
5. **M. S. S. Suresh** and A. Narayanan, "System and methods for maximizing camera coverage for exam surveillance system," 202 041 017 327, Reply Filed. Application in amended examination, Apr. 10, 2020.

Journal Articles

1. **M. S. S. Suresh**, S. S. Rangasrinivasan, A. Olszewski, *et al.*, "Trace the scribbles: A text restoration and artifact correction network for child handwriting enhancement," *International Journal on Document Analysis and Recognition (IJDAR)*, Sep. 1, 2026, Published online 1 September 2026. DOI: 10.1007/s10032-026-00618-1.
2. R. Sasidharan, **M. S. S. Suresh**, and R. Manazhy, "Ipgsl: An intelligent pattern generation method for sign language recognition," *SN Computer Science*, vol. 7, no. 6, p. 683, Aug. 10, 2026. DOI: 10.1007/s42979-026-05261-5.
3. S. Rangasrinivasan, **M. S. S. Suresh**, A. Olszewski, S. Setlur, B. Jayaraman, and V. Govindaraju, "Ai-enhanced child handwriting analysis: A framework for the early screening of dyslexia and dysgraphia," *SN Computer Science*, vol. 6, no. 5, pp. 1–26, 2025.
4. S. Renjith, **M. S. S. Suresh**, and R. Manazhy, "An effective skeleton-based approach for multilingual sign language recognition," *Engineering Applications of Artificial Intelligence*, vol. 143, 2025.
5. **M. S. S. Suresh**, V. Menon, and V. Govindaraju, "Adaptive coverage optimization for energy-constrained multicamera surveillance networks," *IEEE Sensors Journal*, vol. 24, no. 21, pp. 35 528–35 537, 2024. DOI: 10.1109/JSEN.2024.3460536.
6. **M. S. S. Suresh** and V. Menon, "A generic and scalable approach to maximize coverage in diverse indoor and outdoor multicamera surveillance scenarios," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 53, no. 2, pp. 1172–1182, 2023. DOI: 10.1109/TSMC.2022.3194209.
7. **M. S. S. Suresh** and V. Menon, "An efficient graph based approach for reducing coverage loss from failed cameras of a surveillance network," *IEEE Sensors Journal*, vol. 22, no. 8, pp. 8155–8163, 2022. DOI: 10.1109/JSEN.2022.3157819.
8. **M. S. S. Suresh**, A. Narayanan, and V. Menon, "Maximizing camera coverage in multicamera surveillance networks," *IEEE Sensors Journal*, vol. 20, no. 17, pp. 10 170–10 178, 2020. DOI: 10.1109/JSEN.2020.2992076.

Conference Proceedings

1. **M. S. S. Suresh**, S. Rangasrinivasan, A. Olszewski, S. Setlur, and V. Govindaraju, "A recognition-aware multimodal framework for actual and intended text transcription of child handwriting," in *International Workshop on Document Analysis Systems (ICDAR2026)*, Springer, 2026, pp. 371–388.
2. **M. S. S. Suresh**, S. Rangasrinivasan, A. Olszewski, S. Setlur, and V. Govindaraju, "Read the scribbles: A contextual htr for child handwriting recognition without linguistic normalization," in *Workshop on Textual Information Processing & Synthesis in the Wild (ICPR 2026)*, Aug. 2026.
3. A. Harindranath, S. M. S, **M. S. S. Suresh**, V. Nambiar, and V. Menon, "Transegnet: A transformer guided framework for enhanced segmentation of infarct core and affected aspects regions for ischemic stroke assessment," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops*, Mar. 2026, pp. 425–433.
4. **M. S. S. Suresh**, M. S. Sony, V. Menon, S. Setlur, and V. Govindaraju, "Contrast-free brain perfusion imaging: A transformer-guided deep learning approach to CBV mapping," in *Workshop on Health Intelligence (W3PHIAI-26), AAAI 2026*, In press, Jan. 2026.

5. S. Rangasrinivasan, **M. S. S. Suresh**, S. Setlur, B. Jayaraman, and V. Govindaraju, "From scribbles to text: A novel transformer-based recognition model for child handwriting," in *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR)*, 2025, pp. 115–131.

6. **M. S. S. Suresh**, V. Menon, S. Setlur, and V. Govindaraju, "Maximizing coverage over a surveillance region using a specific number of cameras," in *Pattern Recognition*, Cham: Springer Nature Switzerland, 2025, pp. 303–320, ISBN: 978-3-031-78312-8. DOI: 10.1007/978-3-031-78312-8_20.

7. M. S. Sony, **M. S. S. Suresh**, and V. Menon, "Contrast-free ischemic stroke assessment: Generating cbv maps from ncct scans using deep learning models," in *Proceedings of the 2nd International Workshop on Digital and Mobile Smart Health Systems (DM-SMARTHEALTH 2024) in conjunction with the 11th International Conference on Smart Computing (SMARTCOMP 2025)*, Jun. 2025, pp. 1–6.

8. S. Renjith, R. Manazhy, and **M. S. S. Suresh**, "Recognition of sign language using hybrid cnn–rnn model," in *Innovative Computing and Communications*, Singapore: Springer Nature Singapore, 2024, pp. 17–28, ISBN: 978-981-97-3591-4. DOI: 10.1007/978-981-97-3591-4_2.

9. S. Renjith, R. Manazhy, and **M. S. S. Suresh**, "Sign language recognition by using spatio-temporal features," 5th International Conference on Innovative Data Communication Technologies and Application (ICIDCA 2024), vol. 233, 2024, pp. 353–362. DOI: j.procs.2024.03.225.

10. S. Renjith, R. Manazhy, and **M. S. S. Suresh**, "Sign language recognition using bilstm model," in *2024 IEEE 9th International Conference for Convergence in Technology (I2CT)*, 2024, pp. 1–5. DOI: 10.1109/I2CT61223.2024.10543543.

11. S. Renjith, R. Manazhy, and **M. S. S. Suresh**, "Sign language recognition using lstm model: A comparative analysis of csl and arsl datasets," in *Innovative Computing and Communications*, Singapore: Springer Nature Singapore, 2024, pp. 359–368, ISBN: 978-981-97-3591-4. DOI: 10.1007/978-981-97-3591-4_29.

12. M. Sony, **M. S. S. Suresh**, V. Menon, P. Karthika Rani, and V. Nambiar, "Deep learning based generation of cerebral blood flow ct perfusion images from non-contrast ct scans of ischemic stroke patients," in *2024 IEEE 21st India Council International Conference (INDICON)*, 2024, pp. 1–6. DOI: 10.1109/INDICON63790.2024.10958334.

13. **M. S. S. Suresh** and A. Narayanan, "Improving classification accuracy using combined filter+wrapper feature selection technique," in *2019 IEEE International Conference on Electrical, Computer and Communication Technologies (ICECCT)*, 2019, pp. 1–6. DOI: 10.1109/ICECCT.2019.8869518.

Areas of Research

AI for Education and Learning Disabilities, Handwriting Analysis and Document Intelligence

Computer Vision, Image Processing, and Pattern Recognition

Natural Language Processing (NLP), Vision-Language Models, Large Language Models (LLMs).

Multi-Camera Surveillance Systems and Coverage Optimization, Healthcare AI,

Professional Highlights

Publications & Patents	Author or co-author of 20 peer-reviewed publications in leading journals and conferences, including IEEE TSMC, IEEE Sensors Journal, IJDAR, EAAI, and ICDAR; inventor or co-inventor on five Indian patent applications.
Editorial Board Member	Humanities and Social Sciences Communications, Springer Nature (October 2026–Present).
Invited Reviewer	Invited reviewer for journals published by Elsevier, IEEE, Springer, Wiley, and Sage, and for the British Machine Vision Conference (BMVC).